Arquitectura de Robots

The Geometry of Robots

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Section 1.1

Rigid motions

Classification of Robotic Manipulators

1 Rigid motions

The geometry of three-dimensional space and rigid motions plays a central role in all aspects of robotic manipulation.

First, we begin by examining representations of points and vectors in a Euclidean space equipped with multiple coordinate frames.

Introduce the concept of a rotation matrix to represent relative orientations among the coordinate frames.

It can be used to simultaneously represent the position and orientation of one coordinate frame relative to another.

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Representing Positions

The **Synthetic Approach**: Reasons directly related to geometric entities (e.g., points or lines)

The **Analytic Approach** : These entities are depicted through coordinates or equations, and reasoning is carried out through algebraic operations.

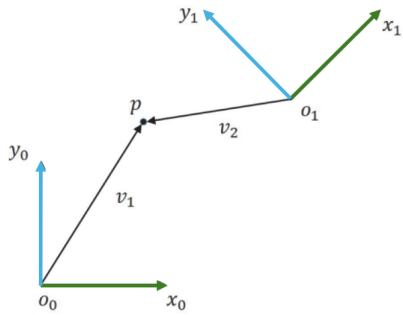
Synthetic approach

2 Representing Positions

Two coordinate frames that differ in orientation by an angle of 45°.

Using the **synthetic approach**, without ever assigning coordinates to points or vectors, one can say that x_0 is perpendicular to y_0, or that $v_1 \times v_2$ defines a vector **p** that is perpendicular to the plane containing v_1 and v_2, in this case pointing out of the page.

Figura 1: *Two coordinate frames.*



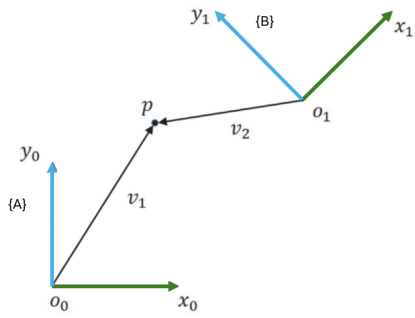
To assign coordinates, it is necessary to specify a reference coordinate frame.

We could specify the coordinates of the point p with respect to the frame $o_0x_0y_0$ or the frame $o_1x_1y_1$.

In this case, we might assign to p the coordinate vector $(5, 6)$ and in the latter case $(-3, 3)$.

$$ {}^{A_0}p = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, {}^{B_1}p = \begin{bmatrix} -3 \\ 3 \end{bmatrix} \qquad (1) $$

Figura 2: *Two coordinate frames A and B rotate 45 degrees.*



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Frame with respect to another

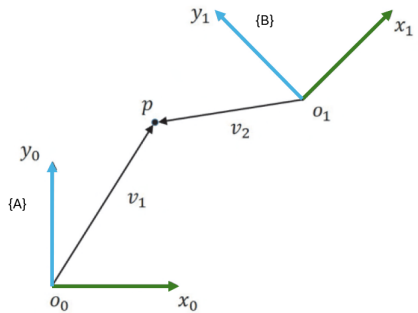
3 Frame with respect to another

This p_0 and p_1 are **coordinate vectors** that represent the location of this **point** in space with respect to coordinate frames $o_0x_0y_0$ and $o_1x_1y_1$, respectively.

Given that the origin of a coordinate system is a point in space, we can allocate coordinates to denote the location of the origin of one coordinate system relative to another.

$$ {}^{A_0}_{B_1}O = \begin{bmatrix} 12 \\ 8 \end{bmatrix}, {}^{B_1}_{A_0}O = \begin{bmatrix} -16 \\ 3 \end{bmatrix} \qquad (2) $$

Figura 3: *Two coordinate frames A and B rotate 45 degrees.*



Frame B with respect to A

3 Frame with respect to another

$$ {}^{A_0}_{B_1}O = \begin{bmatrix} 12 \\ 8 \end{bmatrix} \qquad (3) $$

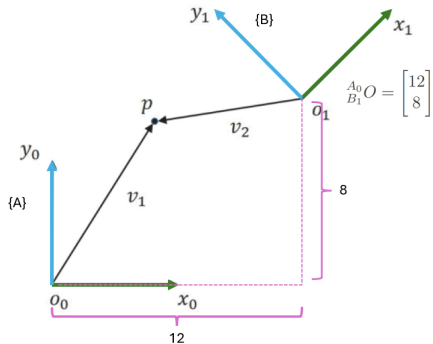


Figura 4: *The Origin of a coordinate B - A rotates 45 degrees..*

Frame A with respect to B

3 Frame with respect to another

$$ {}^{B_1}_{A_0}O = \begin{bmatrix} -16 \\ 3 \end{bmatrix} \qquad (4) $$

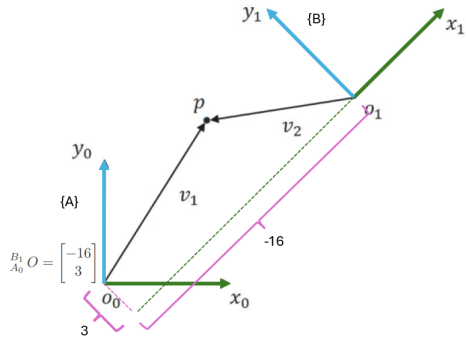


Figura 5: *The Origin of a coordinate A - B rotates 45 degrees.*

Point and Vector

3 Frame with respect to another

- A point corresponds to a specific location in space.
- A vector specifies a direction and a magnitude.
- The point p is not equivalent to the vector v_1
- Points and vectors are not equivalent, since points refer to specific locations in space, but a vector can be moved to any location in space.
- Two vectors are equal if they have the same direction and the same magnitude.

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Assigning coordinates to vectors

v_1 and v_2 are geometric entities that are invariant with respect to the choice of coordinate frames, but the representation by coordinates of these vectors depends directly on the choice of reference coordinate frame.

Example assigning coordinates to vectors

4 Assigning coordinates to vectors

Example:

$$ {}^{A_0}v_1 = \begin{bmatrix} 5 \\ 6 \end{bmatrix} $$

$$ {}^{B_1}v_1 = \begin{bmatrix} 8 \\ 2 \end{bmatrix} $$

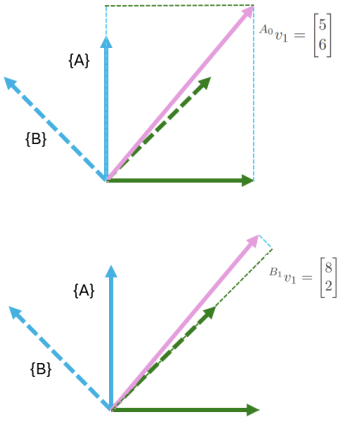


Figura 6: *Example assigning coordinates to vectors.*

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Section 5.1

Descriptions : positions, orientation and frames

Description of a position

A coordinate system is established; we can locate any point in the universe with a 3 x 1 position vector.

$$ {}^{A}P = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \tag{5} $$

This means that the components of ${}^{A}P$ have numerical values that indicate distances along the axes of A.

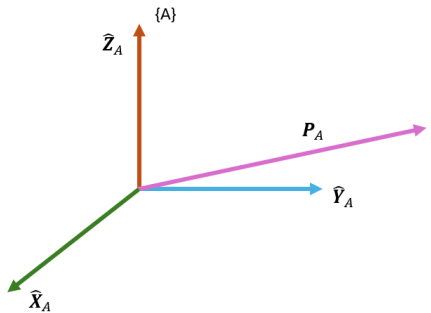


Figura 7: *Vector relative to frame A*

Description of an orientation

We will find it necessary not only to represent a point in space, but also to describe the **orientation** of a body in space.

In order to describe the orientation of a body, we will attach a coordinate system to the body and then give a description of this coordinate system relative to the reference system.

The coordinate system (B) has been attached to the body in a known way.

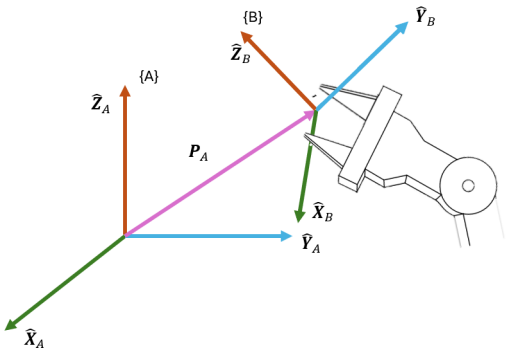


Figura 8: *Locating an object in position and orientation*

Description of a frame

The situation of a **position** and an **orientation** pair arises so often in robotics that we define an entity called a **frame**, which is a set of **four vectors** that give position and orientation information.

One vector locates the fingertip position and three more describe its orientation.

$$\{B\} = \{{}^{A}_{B}R, {}^{A}P_{BORG}\}$$

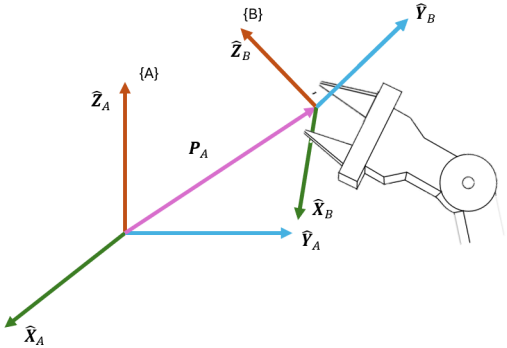


Figura 9: *Locating an object in position and orientation*

Example of several frames

There are three frames that are shown along with the universe coordinate system.

Frames A and B are known relative to the universe coordinate system, and frame C is known relative to frame A.

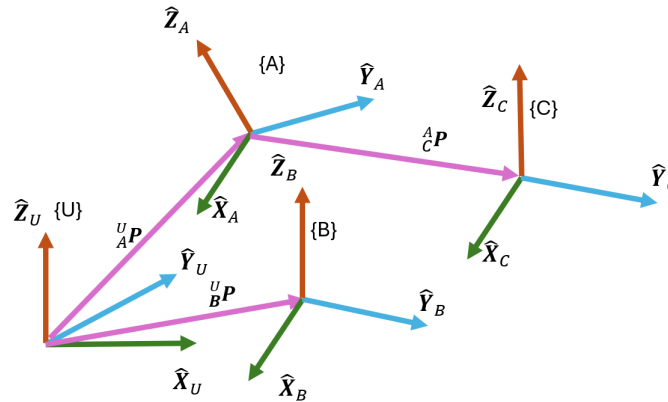


Figura 10: *Example of several frames.*

Changing description from frame to frame translation

We wish to express ${}^{B}P$ this point in space in terms of frame A, when A has the same orientation as B.

In this case, B differs from A only by a translation, which is given by ${}^{A}P_{BORG}$, a vector that locates the origin of B relative to A.

$$ {}^{A}P = {}^{B}P + {}^{A}P_{BORG} \tag{6} $$

Example frame to frame translation

$$^{A}P = {}^{B}P + {}^{A}P_{BORG}$$

$$^{A}P = {}^{B}P + {}^{A}P_{BORG}$$

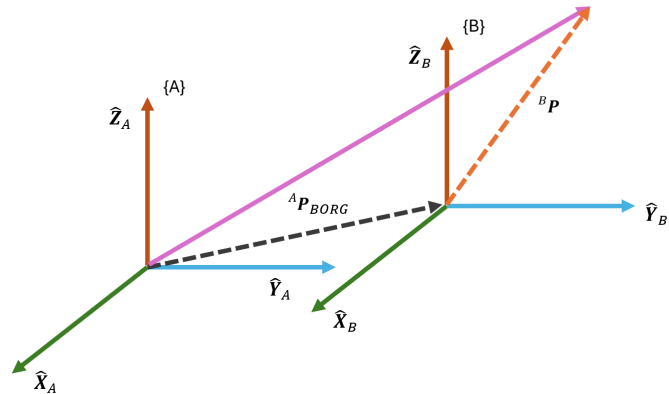


Figura 11: *Example of Translational mapping..*

Section 5.2

Mappings involving rotated frames

Rotation matrix

For convenience, we stack these three unit vectors together as the columns of a 3×3 matrix.

We will call this matrix a rotation matrix, and, if this particular rotation matrix describes B relative to A, we name it with the notation ${}^{A}_{B}R$.

We refer to this matrix as a **rotation matrix**, and since this specific rotation matrix represents B in relation to A

$$ {}^{A}_{B}R = [{}^{A}_{B}X \quad {}^{A}_{B}Y \quad {}^{A}_{B}Z] = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \tag{7} $$

Each component of ${}^A_B R$

Hence, each component of ${}^A_B R$ in (9) can be written as the dot product of a pair of unit vectors:

$$ {}^A_B R = [{}^A_B X \, {}^A_B Y \, {}^A_B Z] = \begin{bmatrix} {}_B X \cdot_A X & {}_B Y \cdot_A X & {}_B Z \cdot_A X \\ {}_B X \cdot_A Y & {}_B Y \cdot_A Y & {}_B Z \cdot_A Y \\ {}_B X \cdot_A Z & {}_B Y \cdot_A Z & {}_B Z \cdot_A Z \end{bmatrix} \tag{8} $$

The dot product is ${}_B X \cdot_A X \cos\theta_{AB}$

Properties of rotation matrix

5 Descriptions : positions, orientation and frames

The rows of the matrix are the unit vectors of A expressed in B; that is,

$$ {}^A_B R = [{}^A_B X \quad {}^A_B Y \quad {}^A_B Z] = \begin{bmatrix} {}^B_A X^T \\ \\ {}^B_A Y^T \\ \\ {}^B_A Z^T \end{bmatrix} \tag{9} $$

${}^B_A R$, the description of frame A relative to B, is given by the transpose of

$$ {}^B_A R = {}^A_B R^T \tag{10} $$

we know that the **inverse of a matrix with orthonormal columns is equal to its transpose**.

Properties of rotation matrix cont...

The columns of a rotation matrix all have unit magnitude, and, further, these unit vectors are orthogonal.

$$ {}^{A}_{B}R = {}^{B}_{A}R^{-1} = {}^{B}_{A}R^{T} \tag{11} $$

Rotation relative to another frame

Frequently, we encounter scenarios where a vector is defined in one frame, B, and we need to determine its definition in another frame, A, given that the origins of both frames coincide.

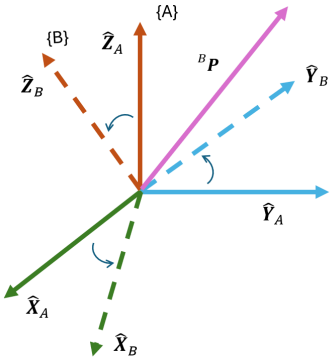


Figura 12: *Rotating the description of a vector.*

Orientation

This orientation is given by the rotation matrix ${}^A_B R$, whose columns are the unit vectors of B written in A.

In order to calculate ${}^A P$, we observe that the components of any vector are merely the projections of that vector onto the unit vectors of its coordinate system. The projection is calculated as **the vector dot product**.

$$
\begin{aligned}
{}^A p_x &= {}^A_B X \cdot {}^B P, \\
{}^A p_y &= {}^A_B Y \cdot {}^B P, \\
{}^A p_z &= {}^A_B Z \cdot {}^B P,
\end{aligned}
\tag{12}
$$

Orientation cont ···

Written compactly, by using a rotation matrix, as

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P \tag{13} $$

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Example : Frame B rotated 30 degrees

6 Example

The figure 39 shows a frame B that is rotated relative to frame A about **Z** by 30 degrees. Here, **Z** is pointing out of the page. Given ${}^{B}P = [0, 2, 0]^{T}$

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

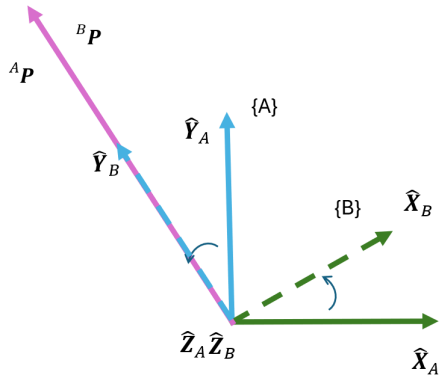


Figura 13: *B rotated 30 degrees.*

Example : Frame B rotated 30 degrees Solution

6 Example

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

$$ {}^{A}_{B}R = \begin{bmatrix} {}_{B}X \cdot_{A} X & {}_{B}Y \cdot_{A} X & {}_{B}Z \cdot_{A} X \\ {}_{B}X \cdot_{A} Y & {}_{B}Y \cdot_{A} Y & {}_{B}Z \cdot_{A} Y \\ {}_{B}X \cdot_{A} Z & {}_{B}Y \cdot_{A} Z & {}_{B}Z \cdot_{A} Z \end{bmatrix} $$

$$ {}_{B}X \cdot_{A} X = $$

$$ \|{}_{B}X\| \cdot \|{}_{A}X\| \cos\theta = $$

$$ 1 \cdot 1 \cdot \cos 30 = \cos 30 $$

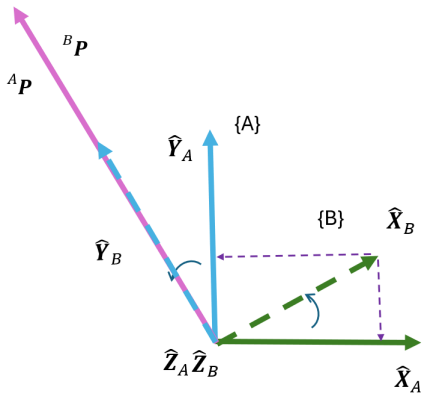


Figura 14: *Vector projection Xb from Xa and Ya.*

Example : Frame B rotated 30 degrees Solution

6 Example

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

$$ {}^{A}_{B}R = \begin{bmatrix} \cos 30 & {}_{B}Y \cdot_{A} X & {}_{B}Z \cdot_{A} X \\ {}_{B}X \cdot_{A} Y & {}_{B}Y \cdot_{A} Y & {}_{B}Z \cdot_{A} Y \\ {}_{B}X \cdot_{A} Z & {}_{B}Y \cdot_{A} Z & {}_{B}Z \cdot_{A} Z \end{bmatrix} $$

$$ {}_{B}X \cdot_{A} Y = $$

$$ \|{}_{B}X\| \cdot \|{}_{A}Y\| \cos\theta = $$

$$ 1 \cdot 1 \cdot (90 - \cos 30) = \sin 30 $$

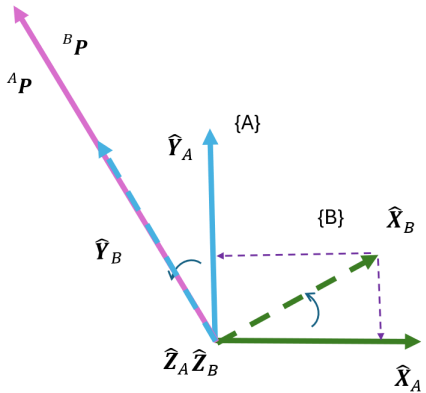


Figura 15: *Vector projection Xb from Xa and Ya.*

Example : Frame B rotated 30 degrees Solution

6 Example

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

$$ {}^{A}_{B}R = \begin{bmatrix} \cos 30 & {}_{B}Y \cdot_{A} X & {}_{B}Z \cdot_{A} X \\ \sin 30 & {}_{B}Y \cdot_{A} Y & {}_{B}Z \cdot_{A} Y \\ {}_{B}X \cdot_{A} Z & {}_{B}Y \cdot_{A} Z & {}_{B}Z \cdot_{A} Z \end{bmatrix} $$

$$ {}_{B}X \cdot_{A} Z = $$

$$ \|{}_{B}X\| \cdot \|{}_{A}Z\| \cos\theta = $$

$$ 1 \cdot 1 \cdot \cos(90) = 0 $$

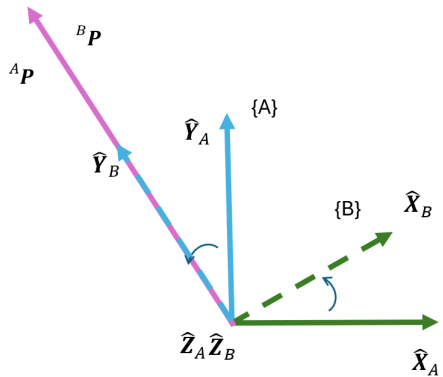


Figura 16: *B rotated 30 degrees.*

Example : Frame B rotated 30 degrees Solution

6 Example

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

$$ {}^{A}_{B}R = \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & {}_{B}Z \cdot {}_{A}Z \end{bmatrix} $$

$$ {}_{B}Z \cdot {}_{A}Z = $$

$$ \|{}_{B}Z\| \cdot \|{}_{A}Z\| \cos\theta = $$

$$ 1 \cdot 1 \cdot \cos(0) = 1 $$

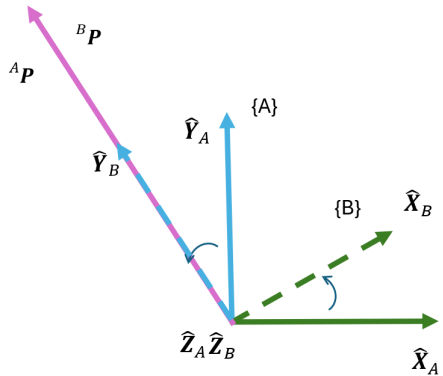


Figura 17: *B rotated 30 degrees.*

Example : Frame B rotated 30 degrees Solution

6 Example

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

$$ {}^{A}_{B}R = \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{A}P = \begin{bmatrix} \cos 30 & -\sin 30 & 0 \\ \sin 30 & \cos 30 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} $$

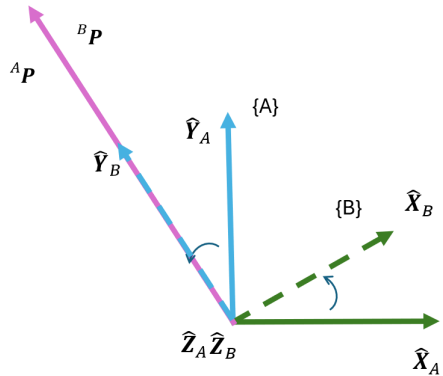


Figura 18: *B rotated 30 degrees.*

Example : Frame B rotated 30 degrees Solution

6 Example

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P $$

$$ {}^{A}P = \begin{bmatrix} 0.866 & -0.500 & 0 \\ 0.500 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = $$

$$ {}^{A}P = {}^{A}_{B}R \ {}^{B}P = \begin{bmatrix} -1 \\ 1.732 \\ 0 \end{bmatrix} $$

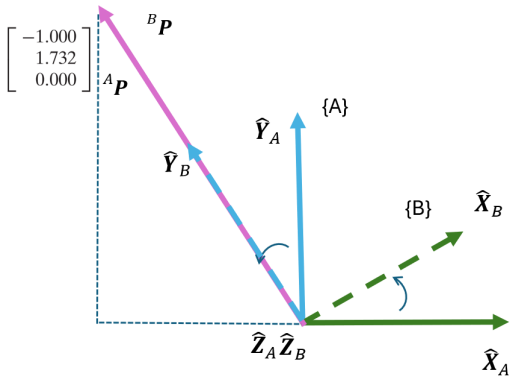


Figura 19: *Vector ${}^{A}P$ Solution .*

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Q&A

Thank you for your attention

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